

HW 1: Basic Python Programming

CPE232 Data Models

Description : For this homework, please write your code in the section we prepared for you. If you need to leave a comment or message for the TA, use the # symbol in the code section. You may also create a text cell in the Jupyter Notebook to provide clarification.

Good luck with Python!



1. Basic usage

John Doe is a 35 years-old software engineer who earns \$35000.00 a month.

Create and assign variables to store this person's information (name, age, position and salary).

```
In [1]: # Write your code here
name = "John Doe"
age = 35
position = "software engineer"
salary = 35000.50
```

What is the type of each variables?

```
In [41]: # Write your code here
print(type(name))
print(type(age))
```

```
print(type(position))
print(type(salary))
```

```
<class 'str'>
<class 'int'>
<class 'str'>
<class 'float'>
```

The manager decides to give John a 7% raise. Update his salary.

```
In [3]: # Write your code here
salary = salary * 1.07
```

Prints his information again with his new salary.

```
In [4]: # Write your code here
print(name)
print(age)
print(position)
print(salary)
```

```
John Doe
35
software engineer
3745053.5
```

2. Variable and Expression

2.1 Write a code to convert temperature unit from celcius to other units and then prints out

```
In [5]: C = 34.5
```

Fahrenheit

$$\frac{C}{5} = \frac{F-32}{9}$$

```
In [6]: F = (C/5) * 9 + 32
print(F)
```

```
94.1
```

Kelvin

$$K = C + 273.15$$

```
In [7]: K = C + 273.15
print(K)
```

```
307.65
```

Rømer

$$Ro = \frac{C \times 21}{40} + 7.5$$

```
In [8]: Ro = C*21/40 + 7.5
        print(Ro)
```

25.6125

2.2 Write code to read the input for weight (kg) and height (cm), then print out the BMI (Body Mass Index).

$$BMI = \frac{kg}{m^2}$$

```
In [10]: weight = int(input("Weight (kg):"))
        height = int(input("Height (cm):"))
        # Write your code here
        BMI = weight/((height/100)**2)
        print(BMI)
```

22.22222222222222

3. Multi-item variables

List

```
In [43]: names = ['Thomas', 'Kate', 'Mike', 'Amelia', 'James', 'Megan']
```

Create new variable call `new_name` which takes input name of the user.

```
In [44]: new_name = input('Enter your name: ')
```

Insert `new_name` into `names` list.

```
In [45]: # Write your code here
        names.append(new_name)
```

Print your name from the list `by index`.

```
In [46]: # Write your code here
        print(names[-1])
```

Chanon

Merge `another_names` into `names`.

```
In [47]: another_names = ['Peter', 'Steve', 'Sam', 'Charlotte']
```

```
In [48]: # Write your code here
        names = names + another_names
        print(names)
```

['Thomas', 'Kate', 'Mike', 'Amelia', 'James', 'Megan', 'Chanon', 'Peter', 'Steve', 'Sam', 'Charlotte']

Change `Amelia`'s name to `Amy`

```
In [49]: # Write your code here
        idx = 0
        for i in range(len(names)) :
```

```

    if names[i] == "Amelia" :
        names[i] = "Amy"
        idx = i
        break
print(names[idx])

```

Amy

Dictionary

```

In [51]: capital_city = {'England': 'London',
                        'Spain': 'Madrid',
                        'Japan': 'Tokyo',
                        'Australia': 'Sydney',
                        'Germany': 'Berlin',
                        }

```

Add a record `Thailand` and it's capital city to this dictionary

```

In [52]: # Write your code here
capital_city.update({'Thailand' : 'Bangkok'})
print(capital_city)

```

```

{'England': 'London', 'Spain': 'Madrid', 'Japan': 'Tokyo', 'Australia': 'Sydney',
'Germany': 'Berlin', 'Thailand': 'Bangkok'}

```

You may notice that the capital city of `Australia` is wrong. It should be `Canberra` .
Correct this mistake.

```

In [53]: # Write your code here
capital_city.update({"Australia" : "Canberra"})
print(capital_city)

```

```

{'England': 'London', 'Spain': 'Madrid', 'Japan': 'Tokyo', 'Australia': 'Canberra',
'Germany': 'Berlin', 'Thailand': 'Bangkok'}

```

4. Control Flows and conditional statements

if...elif...else

1. Write a program that takes two numbers as input from the user, called A and B. Then, print the number that is greater.

Example:

- If `A = 25` and `B = 15` , the program should print: "A = 25"
- If `A = 10` and `B = 22` , the program should print: "B = 22"

```

In [21]: A = int(input("A: "))
B = int(input("B: "))
# Write your code here
if A == 25 and B == 15 :
    print("A = 25")
elif A == 10 and B == 22 :
    print("B = 22")

```

A = 25

2. Define a variable to get input age from user

```
In [ ]: age = int(input("Age: "))
```

25

Write a series of if...elif...else statement that categorize input age into following groups:

- Babies: 0-2 years old
- Children: 3-12 years old
- Teenager: 13-19 years old
- Young Adults: 20-29 years old
- Middle-aged Adults: 30-45 years old
- Old Adult: 46-59 years old
- Elderly: Above 60 years old

```
In [23]: # Write your code here
if age >= 0 and age <= 2 :
    print("Babies")
elif age >= 3 and age <= 12 :
    print("Children")
elif age >= 13 and age <= 19 :
    print("Teenager")
elif age >= 20 and age <= 29 :
    print("Young")
elif age >= 30 and age <= 45 :
    print("Middle-aged Adults")
elif age >= 46 and age <= 59 :
    print("Old Adult")
else :
    print("Elderly")
```

Young

Looping

1. Write a code to create a multiplication table of an input number (multiplier from 1-12).

```
In [54]: # Write your code here
mul = int(input("Multipiler: "))
for i in range(1,13) :
    print((i)*mul)
```

```

12
24
36
48
60
72
84
96
108
120
132
144

```

2. Write a code that construct the following pattern.

```

input: 5
output:
*
**
***
****
*****

```

```

In [25]: # Write your code here
n = int(input())
print(f"input: {n}")
print("output:")
for i in range(1,n+1) :
    print("*" * (i))

```

```

input: 5
output:
*
**
***
****
*****

```

3. Creates a loop to print I love <programming language>! except for Assembly, print Not you, Assembly .

```

In [26]: languages = ['C/C++', 'Python', 'R', 'Java', 'SQLs', 'Assembly', 'Go', 'Rust', '

```

```

In [27]: # Write your code here
for lang in languages :
    if lang == "Assembly" :
        print("Not you, Assembly")
    else :
        print(f"I love {lang}")

```

I love C/C++
 I love Python
 I love R
 I love Java
 I love SQLs
 Not you, Assembly
 I love Go
 I love Rust
 I love Kotlin

4. Write a code to print every number from 1 to 25 except the one that is divisible by 3.

```
In [28]: # Write your code here
for i in range(1,26) :
    if i % 3 != 0 :
        print(i)
```

1
 2
 4
 5
 7
 8
 10
 11
 13
 14
 16
 17
 19
 20
 22
 23
 25

5. Write a code that finds the number that is divisible by 7 in a given range.

```
In [29]: lower_bound = 1
upper_bound = 100
divisor = 7

result = []
```

```
In [30]: # Write your code here
for i in range(lower_bound,upper_bound + 1) :
    if i % 7 == 0 :
        result.append(i)
print(result)
```

[7, 14, 21, 28, 35, 42, 49, 56, 63, 70, 77, 84, 91, 98]

6. Write a code that construct the following pattern.

input: 5
 output:
 *#####
 **#####
 ***####

```
****##  
*****#  
  
input: 10  
output:  
*#####  
**#####  
***#####  
****#####  
*****#####  
*****#####  
*****###  
*****###  
*****###  
*****###  
*****###
```

```
In [31]: # Write your code here
n = int(input())
print(f"input: {n}")
print("output:")
for i in range(1,n+1) :
    print("'" * i + "#" * (n-i+1))
```

```
input: 10  
output:  
*#####  
**#####  
***#####  
****#####  
*****#####  
******#####  
*******#####  
*****#####  
****#####  
***#####  
**#####  
*#####
```

5. Functions

1. Define a function `average` that takes *list of numbers* and calculate the mean of input. It should look like this:

average([1,2,3,4]) output: 2.5

```
In [57]: # Write your code here
def average(list) :
    total = 0
    for num in list :
        total += num
    print(f"output: {total/len(list)}")
```

```
In [58]: average([1,2,3,4])
```

```
output: 2.5
```

2. Define a function `sumproduct` that takes 2 *equal-sized* lists and calculate sum of the products of two lists.

It should look like this:

```
sumproduct([1,2,3],[4,5,6])
output: 32
```

$$(1 * 4) + (2 * 5) + (3 * 6) = 32$$

```
In [55]: # Write your code here
def sumproduct(list1,list2) :
    ans = 0
    for i in range(len(list1)) :
        ans += (list1[i] * list2[i])

    print(f"output {ans}")
```

```
In [56]: sumproduct([1,2,3],[4,5,6])
```

output 32

3. Define a function `fibonacci` that returns Fibonacci number at `n` position.

A Fibonacci number at position `n` is defined by $F(n) = F(n-1) + F(n-2)$. Where

$F(0) = 0$ and $F(1) = 1$

Example: 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, ...

```
In [65]: # Write your code here
n = int(input())
dp = {}

def fibo(n) :
    if n in dp :
        return dp[n]
    if n == 2 :
        return 1
    if n == 1 :
        return 0
    dp[n] = fibo(n-1) + fibo(n-2)
    return dp[n]

print(fibo(n))
```

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4. Define a function `is_palindrome` that takes input string and check whether it is a palindrome or not.

A string is a palindrome if it reads the same forward and backwards.

Example: madam , race car , borrow or rob , amore roma , never odd or even

Do not consider whitespace. Use `str.replace(' ', '')` to remove whitespace from your string.

Case-insensitive. You can turn everything into lower or uppercase using `str.lower()` or `str.upper()`

Hint: you can reverse the string using `[::-1]` slice.

```
In [66]: str1 = "radar" # palindrome
         str2 = "rotator" # palindrome
         str3 = "lemon" # not palindrome
```

```
In [67]: # Write your code here
def is_palindrome(str) :
    str.replace(' ', '')
    str.lower()
    re_str = str[::-1]
    if re_str == str :
        print("palindrome")
    else :
        print("not palindrome")

is_palindrome(str1)
is_palindrome(str2)
is_palindrome(str3)
```

```
palindrome
palindrome
not palindrome
```

5. An **anagram** is a word or phrase formed by rearranging the letters of a different word or phrase.

Define a function **is_anagram** that takes in 2 strings and check whether it is possible to compose a second string using letters in the first string or not.

Example: Tom Marvolo Riddle can be rearranged into I am Lord Voldemort
 Meaning of Life can be rearranged into Engine of a Film

Do not consider whitespace. Use `str.replace(' ', '')` to remove whitespace from your string.

Case-insensitive. You can turn everything into lower or uppercase using `str.lower()` or `str.upper()`

Returns only **True** or **False**

```
In [70]: # Write your code here
str1 = "Meaning of Life"
str2 = "Engine of a Film"

def is_anagram(str1, str2) :
    freq1 = getFreq(str1)
    freq2 = getFreq(str2)
    if freq1 == freq2 :
        return True
    else :
        return False

def getFreq(str) :
    str = str.replace(' ', '').lower()
    freq = {}
    for i in str :
        if i in freq :
            freq[i] += 1
        else :
            freq[i] = 1
    return freq
```

```
In [71]: print(is_anagram(str1, str2))
```

True
